

An Instrumented Edge Node for Dual-Sensor Maritime Safety Monitoring

Carmine Coppola, Vincenzo Bucciero, Simone Perrotta, and Raffaele Montella
Department of Science and Technology, University of Naples “Parthenope”, Naples, Italy
{carmine.coppola001, vincenzo.bucciero001, simone.perrotta001}@studenti.uniparthenope.it
raffaele.montella@uniparthenope.it

Abstract—Maritime safety monitoring benefits from low-cost sensing platforms that can operate near the point of observation, but embedded vision pipelines are often reduced to a single detector benchmark and rarely characterized as complete, measurable systems. This poster paper presents an edge-computing node that combines a dual-view RGB camera assembly with an on-demand thermal sensor behind a common, model-independent processing path, targeting small-vessel and coastal safety monitoring. A Raspberry Pi 4 prototype integrates ONNX-based inference, REST services, and a browser dashboard. Both RGB feeds sustained 10 FPS at 53.16% mean CPU utilization (one logical core) and approximately 450 MiB resident memory over 151 steady-state observations; over 50 replay-based requests the instrumented pipeline required 1.08 s on average and 0.94 s at the median, after stage-level instrumentation had cut that latency by approximately 72%, from 3.32 s, without changing the detection backend. A 120 s full-core stress test peaked at 77.4 °C against a 78 °C cutoff without throttling, and a two-hour run of concurrent acquisition sustained two RGB views at about 9.55 FPS alongside a persistent thermal stream at 9.11 FPS with no invalid frame and no restart. The RGB and thermal channels are heterogeneous, not hardware-synchronized or geometrically registered; the contribution is a validated, instrumented edge architecture and a resource/latency characterization, not a fused multimodal detector or a certified safety system.

Index Terms—maritime safety, edge computing, embedded vision, thermal imaging, object detection, ONNX, resource-constrained inference

I. Motivation

Small vessels and coastal sites are hard to cover with fixed infrastructure, yet they are where early visual warning of people, small craft or floating hazards is most useful. Cameras alone are unreliable under glare, low light and long viewing distances; thermal sensing retains contrast where RGB degrades, but a detector benchmark is not a deployable system. A field-usable node must also manage heterogeneous sensors, run inference inside a tight resource envelope, expose its state, and keep latency operationally meaningful.

II. System Architecture

The platform separates hardware access, source supervision, frame delivery, inference, and presentation through stable interfaces (Fig. 1). A System Orchestrator resolves dependencies and aggregates readiness across services. A Device Manager tracks physical sensors independently of a Source Manager that exposes live cameras, recorded video,

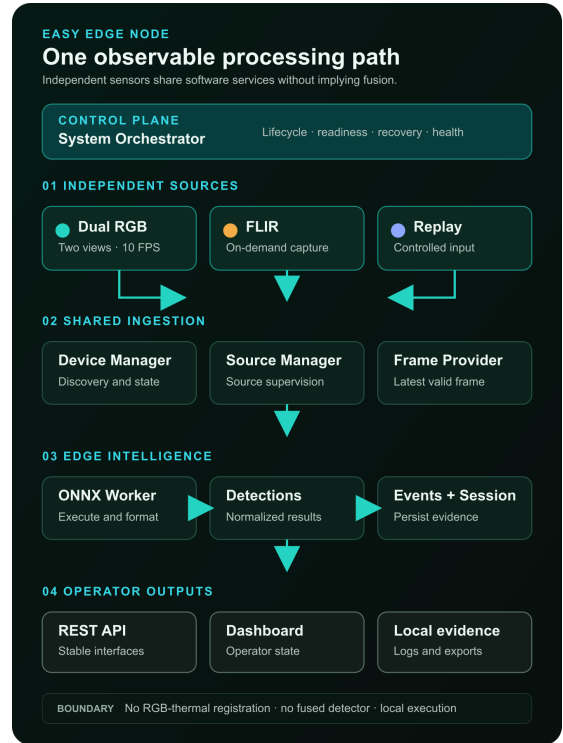


Fig. 1: Poster-specific overview of the EASY runtime. Independent RGB, on-demand thermal, and replay sources share one observable edge-processing path; the prototype does not perform RGB–thermal registration or sensor fusion.

and replay sessions through one common abstraction. A Unified Frame Provider returns the latest valid frame regardless of source type, so live acquisition and replay traverse the same downstream path: inference, detection normalization, event propagation, session association, persistence, and dashboard delivery. Replay inputs therefore exercise the identical code path as live sensors, so processing changes can be tested before reaching the field.

An Inference Worker performs preprocessing and executes a configured ONNX model, adapting backend-specific output into a normalized detection form so the platform is not tied to one detector. RGB and thermal acquisition are decoupled from inference: cameras continually publish frames while selected frames are processed on request, and the on-demand thermal sensor is activated only when needed rather than streamed continuously. The

TABLE I: Key measurements from the integrated prototype.

Metric	Value
Steady-state RGB frame rate (both feeds)	10 FPS
Mean process CPU (one logical core = 100%)	53.16%
Mean process-tree resident memory	450.36 MiB
Replay latency, mean/median/P95 ($n=50$)	1.08 / 0.94 / 1.68 s
Model backend share of one request	588 ms
Peak CPU temperature, 120 s full-core load	77.4 °C
Automated tests passed	310 / 310

two modalities are acquired and exposed through the same source abstraction but are not hardware-synchronized or geometrically registered in the current prototype—they constitute heterogeneous sensing rather than a fused multimodal detector.

III. Prototype and Evaluation

The prototype runs on a Raspberry Pi 4 (8 GB RAM, 64-bit Linux) with an Arducam stereo assembly delivering one 1280×480 MJPEG stream at 10 FPS split into two logical RGB views, a FLIR Lepton thermal module accessed on demand, Flask-based REST services, and a CPU-executed ONNX detector. The detector is a YOLOv8n model (FP32, 640×640 input) trained on a 3-class maritime set (boat, ship, buoy) from SMD and SeaShips. The model was trained under a split that keeps every video sequence inside one subset. All 310 automated tests passed (150 service-level, 160 interface-level), alongside scripted hardware and browser checks covering device registration, dual-view acquisition, thermal capture, and replay.

Over 151 steady-state observations, both logical RGB feeds sustained 10 FPS while the process tree used 53.16% of one logical core and approximately 450 MiB of resident memory (Table I). Fifty replay-based inference requests completed with a mean end-to-end latency of 1.08 s and a median of 0.94 s; the mean is raised by a few slow persistence writes, so both are reported. Decomposing the pipeline into acquisition, preprocessing, backend execution, postprocessing, preview generation, and persistence stages had earlier cut mean latency from 3.32 s to 0.91 s, about 72%, without altering the detection backend—persistence and request coordination, not the model, were the dominant avoidable cost. Model execution accounts for 588 ms of a request and storage for 290 ms. Two robustness tests bound the envelope: 120 s of full-core load reached 77.4 °C against the 78 °C cutoff without throttling, both RGB feeds still at 10 FPS, leaving 0.6 °C of margin; and two hours of continuous inference completed 2,530 requests without error, mean latency rising to 2.34 s under that saturated cycle. On the thermal path, eight on-demand FLIR captures with both RGB feeds active completed in 0.89 ± 0.39 s (range 0.65–1.77 s). The operator dashboard exposes live and replayed sources, detections and system state for on-site monitoring.

Concurrent acquisition. A candidate persistent thermal worker, behind a feature flag and not enabled on the deployed unit, ran for two hours with both RGB views:

64,801 valid Y16 frames at 9.11 FPS, RGB at about 9.55 FPS each, no invalid frame, no restart, 56.0–63.3 °C. The paths coexist; synchronized exposure, calibration and fusion stay unvalidated.

Transfer to unseen data. With each video sequence confined to one subset the model reaches mAP50 0.627, against 0.942 under an earlier split that shared sequences; on unseen public near-shore video a diagnostic pilot measured 1.69% vessel AP50. Labelled field data is the next step.

IV. Discussion and Future Work

Dual-view RGB acquisition and thermal sensing fit the envelope of a low-cost board while staying measurable end-to-end, and most avoidable latency lived outside the model. The current prototype does not synchronize or geometrically register RGB and thermal frames, does not claim operational detector accuracy for maritime hazard classes, and is not a certified distress, navigation, or vessel-traffic system. The thermal margin measured here is specific to this unit’s passive cooling, so active cooling is a precondition for an enclosed field build. Future work covers timestamp alignment and geometric calibration as preconditions for genuine RGB–thermal fusion, labelled maritime evaluation across sea states, and longer field trials.

V. Conclusion

This poster paper presents an instrumented edge node exposing live and replayed RGB and thermal sources through one common, measurable processing path. Sustained 10 FPS dual-view acquisition inside a 450 MiB footprint, a sub-second median request latency, a measured thermal margin and two hours of concurrent acquisition show that resource-constrained hardware can support responsive, auditable monitoring at the edge—a foundation for future deployments, not a claim of fused perception or safety certification.

Acknowledgment

This work is partially funded by the EASY project (CUP I63C25000510007).

References

- [1] L. A. Varga, B. Kiefer, M. Messmer, and A. Zell, “SeaDronesSee: A maritime benchmark for detecting humans in open water,” in Proc. IEEE/CVF WACV, 2022, pp. 2260–2270.
- [2] M. Lacki, “Determining the level of threat in maritime navigation based on the detection of small floating objects with deep neural networks,” *Sensors*, vol. 24, no. 23, p. 7505, 2024.
- [3] J.-H. Kim, N. Kim, Y.-W. Park, and C.-S. Won, “Object detection and classification based on YOLO-v5 with improved maritime dataset,” *J. Mar. Sci. Eng.*, vol. 10, no. 3, p. 377, 2022.
- [4] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, “Edge computing: Vision and challenges,” *IEEE Internet Things J.*, vol. 3, no. 5, pp. 637–646, 2016.
- [5] Z. Shao, W. Wu, Z. Wang, W. Du, and C. Li, “SeaShips: A precisely annotated dataset for ship detection,” *IEEE Trans. Multimedia*, vol. 20, no. 10, pp. 2593–2604, 2018.
- [6] R. Montella, S. Kosta, and I. Foster, “DYNAMO: Distributed leisure yacht-carried sensor-network for marine data crowdsourcing,” in Proc. IEEE IC2E, 2018, pp. 333–339.